Richard H. Zhong

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Education

New York University, Tandon School of Engineering Master of Science, Expected May 2026
Major: Computer Engineering
GPA: 3.9
Coursework:
Machine Learning, Deep Learning, Reinforcement Learning for Autonomous Systems, High Performance ML

University of Massachusetts Amherst Bachelor of Science, May 2022
Double Majors: Managerial Economics & Computer Science
Awards: Feldman-Vorwerk Family Undergraduate Research Award

Technical Skills

Languages: Python, CUDA, C, C++, R, MATLAB, Java, JavaScript, SQL, Assembly (x86)
Libraries: PyTorch, NumPy, Pandas, Hugging Face (TRL, Transformers, Datasets), vLLM, Unsloth, Weights & Biases
Model Alignment & Post-training: SFT, RLHF/RLVR, DPO, PPO, GRPO
High Performance Computing: Profiling/Tracing, CUDA Kernels, SLURM
Training Efficiency: PEFT/LoRA, FlashAttention, DeepSpeed, FSDP

Research Interests

AI safety with a focus on scalable oversight, preference learning, and mechanistic interpretability. Interested in aligning LLMs to heterogeneous and conflicting human values.

Research Experience

Master's Thesis September 2025 - Present
Advised by Prof. Chinmay Hegde, NYU Tandon
Thesis: Adversarial Rubric Optimization for LLM Alignment

- Building an adversarial framework to automatically identify and patch weaknesses in LLM evaluation rubrics in non-verifiable domains, improving the reliability of AI judges used in alignment pipelines.
- Using GRPO to train exploit models and rubric generating models in an adversarial training loop, with a custom reward function to improve preference consistency in judge models that utilize rubric generators.

Course Research Project March 2025 - May 2025
ECE-GY 9143 High-Performance Machine Learning, NYU Tandon
Topic: Fine-tuning LLMs Without Normalization Layers

- Investigated the training dynamics of replacing Layer Normalization with Dynamic Tanh (DyT) in Pythia models during Supervised Fine-Tuning using RE-WILD SFT data mix.
- Quantified the scaling relationship between parameter count and training stability: DyT incurred ~2× higher final loss at 17M parameters but recovered to within ~1.9× at 410M, suggesting stability is a function of model scale.

Projects

CUDA Kernel Optimization for Memory-Efficient GPU Compute March 2025

- Engineered custom CUDA kernels for vector addition and matrix multiplication; benchmarked the impact of memory coalescing and shared memory tiling on throughput: optimized kernels to achieve a ~2.1× GFLOPS gain over naive implementations.

Performance Optimization of ResNet-18 Training on HPC Systems April 2025

- Profiled training performance of ResNet-18 on CIFAR-10 on HPC V100 nodes using PyTorch tracing; identified 4 DataLoader workers as optimal, reducing I/O overhead by ~4.5×, and quantified batch normalization removal as a ~9% training speedup at the cost of ~7 points accuracy.

Fully Convolutional Network Image Classification on the CBIS-DDSM Dataset March 2022

- Implemented neural network to classify benign and malignant cancer imaging in the Curated Breast Imaging Subset of Digital Database for Screening Mammography.
- Employed a strictly convolutional architecture, eliminating fully connected layers to exploit input-size agnostic properties and accommodate varying mammography dimensions.

Volunteer Experience

Etheros HealthData Foundation, Volunteer, *Boston, MA* May 2024 - November 2025

- Nonprofit volunteering focused on promoting decentralization of health data management. Collaborated with American University researching data privacy vulnerabilities during healthcare transitions.